Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*	(i)	Waves from source interfere (accept superpose) with waves <u>reflected</u> [from plate] (1) Waves in phase **or** constructively interfere at [each] antinode (1)		2		2		
		(ii)	Antinodes $\frac{\lambda}{2}$ apart or λ = 32 mm (1) 8 mm = ¼ λ or equivalent, so node at P (1) If correct answer and valid argument without any reference to λ award 1 mark. Alternative for 1 mark only – phase change of 180^{o} <u>on reflection</u> therefore node at P		1	1	2		
	(b)	(i)	In antiphase. Accept exactly**, or** π**, or** <u>180° out of phase</u> or half a cycle (1) don't accept phases expressed in terms of λ Longest λ is 0.60 m [because 0.30 m = $\frac{\lambda}{2}$] (1) Second longest is 0.20 m [because 0.30 m = $\frac{3\lambda}{2}$] (1)	1	1 1		3		
		(ii)	f = 20 [Hz] **or** T = 0.050 s if <u>used</u> in $v = \frac{\lambda}{T}$ (1) If λ = 0.60 m, then $v = 20 \times 0.6 = 12$ [$\mathrm{m\ s^{-1}}$] (1) **ecf** on both λ This is in range; λ = 0.20 [m] is out of range (1) **Alternative method for last two marks** $10\ \mathrm{m\ s^{-1}}$ at 20 Hz corresponds to λ = 0.50 m, **and** $15\ \mathrm{m\ s^{-1}}$ at 20 Hz corresponds to λ = 0.75 m (1) So λ = 0.60 m (1) **ecf** on (i) extending to conclusion (also rewards strategy)		3		3	2	
			Question 6 total	**1**	**8**	**1**	**10**	**2**	**0**